MACHINE LEARNING PROJECT REPORT

USED CAR PRICE PREDICTION
USING MACHINE LEARNING REGRESSION

An applied Machine Learning workflow for analysing vehicle attributes and predicting the estimated selling price of used cars using regression algorithms.

PROJECT	Used Car Price Prediction
TASK	Regression-Based Price Prediction
SELECTED MODEL	Random Forest Regressor
DATASET	CarDekho Used Car Dataset
TEST SET	1,385 Records (20% Hold-out)
TECHNOLOGIES	Python | Pandas | NumPy | Matplotlib | Seaborn | Scikit-learn

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ABSTRACT

The increasing demand for pre-owned vehicles has created a need for reliable methods to estimate their selling prices. Determining the appropriate price of a used car can be challenging because it depends on several factors, including manufacturing year, kilometres driven, fuel type, transmission, ownership history, mileage, engine capacity, and maximum power.

This project, titled **"Used Car Price Prediction Using Machine Learning Regression,"** aims to develop a machine learning model capable of predicting used car selling prices based on vehicle specifications. The CarDekho used car dataset was used for this purpose. The dataset underwent data cleaning, preprocessing, feature engineering, and exploratory data analysis to identify important patterns and relationships between vehicle characteristics and selling prices.

Four regression algorithms were implemented and evaluated: Linear Regression, Decision Tree Regression, Random Forest Regression, and Tuned Random Forest Regression. Their performance was compared using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the R^2 Score.

Among the evaluated models, the Original Random Forest Regressor achieved the best overall performance, with an MAE of ₹71,381.00, an RMSE of ₹138,354.48, and an R^2 Score of 0.9216 on the test dataset. The selected model was integrated with its preprocessing steps into a single machine learning pipeline and saved using Joblib. Its prediction functionality was successfully tested using new vehicle specifications.

The project demonstrates how machine learning regression techniques can be applied to used car price estimation and provides a foundation for developing an automated vehicle price prediction application.

1. INTRODUCTION

- The automobile industry has experienced significant growth in the pre-owned vehicle market. Used cars have become a popular choice among customers because they are generally more affordable than new vehicles. However, determining the appropriate selling price of a used car is a challenging task, as its value depends on several factors, including manufacturing year, distance travelled, fuel type, transmission, ownership history, mileage, engine capacity, and overall vehicle specifications.
- Traditional methods of estimating used car prices often rely on manual inspection, market knowledge, and comparisons with similar vehicles. These methods may produce inconsistent results because different sellers can assign different prices to vehicles with similar characteristics. Therefore, an automated and data-driven approach can help make the price estimation process more consistent and systematic.
- Machine Learning provides techniques for analysing historical data, identifying patterns, and making predictions based on previously observed information. Regression algorithms are particularly useful for predicting continuous numerical values, such as vehicle selling prices.
- This project focuses on developing a Used Car Price Prediction system using machine learning regression techniques. The CarDekho used car dataset is used to analyse vehicle characteristics and their relationship with selling prices. The dataset is processed through data cleaning, feature engineering, and exploratory data analysis before being used to train different regression models.
- The project implements Linear Regression, Decision Tree Regression, and Random Forest Regression, along with hyperparameter tuning of the Random Forest model. Their performance is evaluated using MAE, RMSE, and R^2 Score to identify the most suitable model.
- The selected model is integrated into a complete machine learning pipeline that combines data preprocessing and price prediction. The trained pipeline is saved using Joblib and tested with new vehicle specifications to demonstrate its prediction capability.
- The overall aim is to demonstrate how machine learning can support used car price estimation through historical data analysis and predictive modelling.

2. PROBLEM STATEMENT

Determining the appropriate selling price of a used car is a challenging task because its value depends on multiple factors, such as manufacturing year, kilometres driven, fuel type, transmission, ownership history, mileage, engine capacity, and maximum power. In the absence of a systematic pricing approach, sellers may set unrealistic prices, while buyers may find it difficult to determine whether a vehicle is fairly valued. Traditional methods of used car price estimation generally depend on manual inspection, personal experience, and comparisons with similar vehicles available in the market. These approaches may lead to inconsistent price estimates because they do not always consider the combined influence of different vehicle specifications. Therefore, there is a need for a data-driven approach that can analyse historical vehicle information and estimate selling prices more systematically. The proposed project addresses this problem by applying machine learning regression algorithms to the CarDekho used car dataset. It aims to identify relationships between vehicle characteristics and selling prices and develop a predictive model that can estimate the selling price of a used car based on its specifications.

3. PROJECT OBJECTIVES

- The primary objective of this project is to develop a machine learning regression model that can predict the selling price of used cars based on their specifications and characteristics. The project aims to analyse historical vehicle data to understand how factors such as manufacturing year, kilometres driven, fuel type, transmission, ownership history, mileage, engine capacity, and maximum power influence vehicle prices.
- Another important objective is to prepare the dataset for machine learning by performing data cleaning, handling missing values, removing duplicate records, and converting vehicle specifications into suitable numerical and categorical formats. Exploratory Data Analysis is conducted to identify important patterns, distributions, correlations, and relationships between vehicle attributes and selling prices.
- The project also aims to implement and compare different regression algorithms to identify a model that provides reliable predictions. Model performance is evaluated using appropriate regression metrics, followed by hyperparameter tuning to investigate whether predictive performance can be improved. Finally, the selected model is combined with its preprocessing operations into a single pipeline, saved using Joblib, and tested with new vehicle specifications to demonstrate its practical application.

4. DATASET DESCRIPTION

- The dataset used in this project is the CarDekho Used Car Dataset, which contains information about pre-owned vehicles and their selling prices. It provides details about different vehicle characteristics, including manufacturing year, kilometres driven, fuel type, seller type, transmission, ownership history, mileage, engine capacity, maximum power, torque, and seating capacity. These attributes provide useful information for analysing the factors that influence the selling price of a used vehicle.
- The original dataset consists of 8,128 records and 13 features. During the initial data inspection, missing values and duplicate records were identified. After removing duplicate records, investigating invalid values, and excluding one vehicle record with an unusually high kilometre reading that was considered potentially erroneous, the final cleaned dataset contained 6,925 records and 15 features. Additional features, such as manufacturer and mileage unit, were extracted during the feature engineering process.
- The target variable selected for prediction is **selling_price**, which represents the selling price of the vehicle. The remaining relevant vehicle attributes are used as input features for training the regression models. The vehicle name column was excluded from model training because it contained a large number of distinct vehicle names, while the manufacturer information was extracted and retained as a separate feature.
- The dataset contains both numerical and categorical attributes. Numerical attributes include manufacturing year, kilometres driven, mileage, engine capacity, maximum power, torque, and seating capacity. Categorical attributes include fuel type, seller type, transmission, ownership category, manufacturer, and mileage unit.
- The dataset was divided into training and testing subsets using an 80:20 ratio. This allowed the machine learning models to learn patterns from the training data and evaluate their predictive performance on previously unseen test data.

5. TOOLS AND TECHNOLOGIES USED

- The project was developed using Python as the primary programming language because of its extensive support for data analysis, data preprocessing, visualization, and machine learning. Jupyter Notebook was used as the development environment to execute code, analyse the dataset, visualize results, and document the complete workflow.
- Pandas and NumPy were used for data manipulation, numerical computations, dataset cleaning, and feature engineering. Matplotlib and Seaborn were used to generate visualizations such as histograms, box plots, scatter plots, bar charts, and correlation heatmaps to understand the distribution of vehicle prices and relationships between different attributes.
- Scikit-learn was the primary machine learning library used for implementing regression algorithms, splitting the dataset into training and testing subsets, handling missing values, scaling numerical features, encoding categorical variables, and evaluating model performance. It also provided RandomizedSearchCV for hyperparameter tuning and Pipeline and ColumnTransformer for integrating preprocessing and model training into a single workflow.
- Joblib was used to save and load the trained machine learning pipeline so that it could be reused for predicting the selling prices of new vehicles without repeating the entire training process.

6. METHODOLOGY

- The proposed system follows a structured machine learning methodology to predict the selling prices of used vehicles based on their specifications. The process begins with loading the CarDekho dataset into a Pandas DataFrame, followed by an initial examination of its dimensions, feature names, data types, missing values, and duplicate records. This initial analysis helps understand the dataset's structure and identify data quality issues that need to be addressed before model development.
- The next stage involves data cleaning and preprocessing. Duplicate records are removed, missing values are identified, and vehicle specifications stored as text are converted into appropriate numerical formats. Feature engineering is performed to extract additional information, such as the vehicle manufacturer and mileage unit. Invalid values, including zero mileage and zero maximum power, are treated as missing values. An unusually high kilometre reading identified during the investigation is also excluded from the cleaned dataset.
- After data preparation, Exploratory Data Analysis is performed to understand the distribution of selling prices and investigate relationships between vehicle characteristics and their prices. Statistical summaries, histograms, box plots, scatter plots, bar charts, and correlation heatmaps are used to identify important patterns and associations within the dataset.
- The cleaned dataset is then divided into input features and the target variable. The selling price is selected as the target, while the relevant vehicle specifications are used as input features. The vehicle name is excluded because of its high number of unique values, while the manufacturer is retained as a separate categorical feature. The dataset is split into training and testing subsets using an 80:20 ratio.
- The preprocessing stage handles numerical and categorical features separately. Missing numerical values are filled using median imputation, followed by feature scaling using StandardScaler. Missing categorical values are handled using most-frequent imputation, and OneHotEncoder converts categorical attributes into numerical representations. These operations are integrated using ColumnTransformer and Pipeline to ensure consistent preprocessing and prevent information leakage from the test dataset into model training.
- Four regression model configurations are subsequently implemented and evaluated: Linear Regression, Decision Tree Regression, Original Random Forest Regression, and Tuned Random Forest Regression. Their performance is compared using Mean Absolute Error, Root Mean Squared Error, and R^2 Score. RandomizedSearchCV with five-fold cross-validation is applied to investigate suitable Random Forest hyperparameters.
- Based on the overall evaluation results, the Original Random Forest Regressor is selected as the final model. Its preprocessing operations and regression algorithm are combined into a single machine learning pipeline and trained using the training dataset. The completed pipeline is saved in Joblib format for future use. Finally, the saved model is loaded and tested using new vehicle specifications to verify that it can generate selling price predictions successfully.

7. EXPLORATORY DATA ANALYSIS (EDA)

7.1 Selling Price Distribution

- The distribution of selling prices was analysed to understand the price range and identify the general pattern of used vehicle prices in the dataset. After data cleaning, the dataset contained 6,925 vehicle records, with selling prices ranging from ₹29,999 to ₹1,00,00,000. The average selling price was approximately ₹5,17,265.95, while the median selling price was ₹4,00,000.
- The mean selling price being higher than the median indicates a positively skewed distribution. This suggests that most vehicles are concentrated within comparatively lower price ranges, while a smaller number of expensive vehicles increase the overall average. The presence of high-priced vehicles also indicates that the dataset contains vehicles belonging to different market segments.

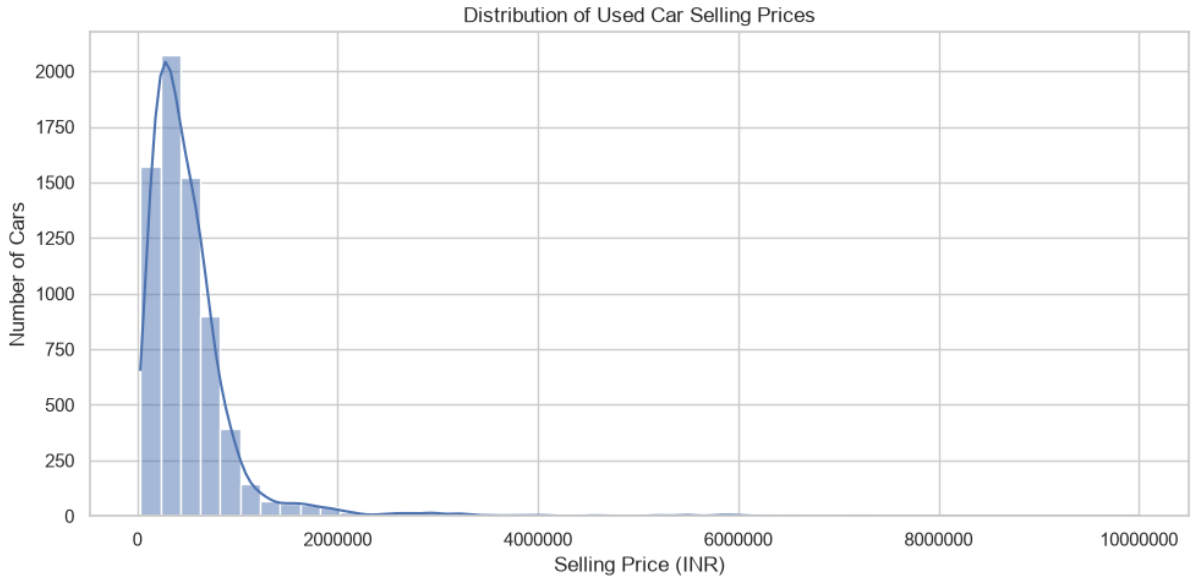


Figure 7.1: Distribution Of Used Car Selling Price

7.2 Manufacturing Year vs Selling Price

- The relationship between manufacturing year and selling price was analysed to understand how vehicle age influences its market value. The analysis indicates a general upward trend in selling prices for newer vehicles. Older vehicles generally have lower selling prices compared to vehicles manufactured in more recent years.
- For example, vehicles manufactured in 2005 had an average selling price of approximately ₹1,37,994, whereas vehicles manufactured in 2015 had an average selling price of approximately ₹5,89,178. Similarly, vehicles manufactured in 2019 recorded an average selling price of approximately ₹9,49,669.
- These observations suggest that manufacturing year is an important factor associated with used car prices. However, the relationship is not perfectly linear because other factors, such as manufacturer, model, engine specifications, and vehicle condition, may also influence the selling price.

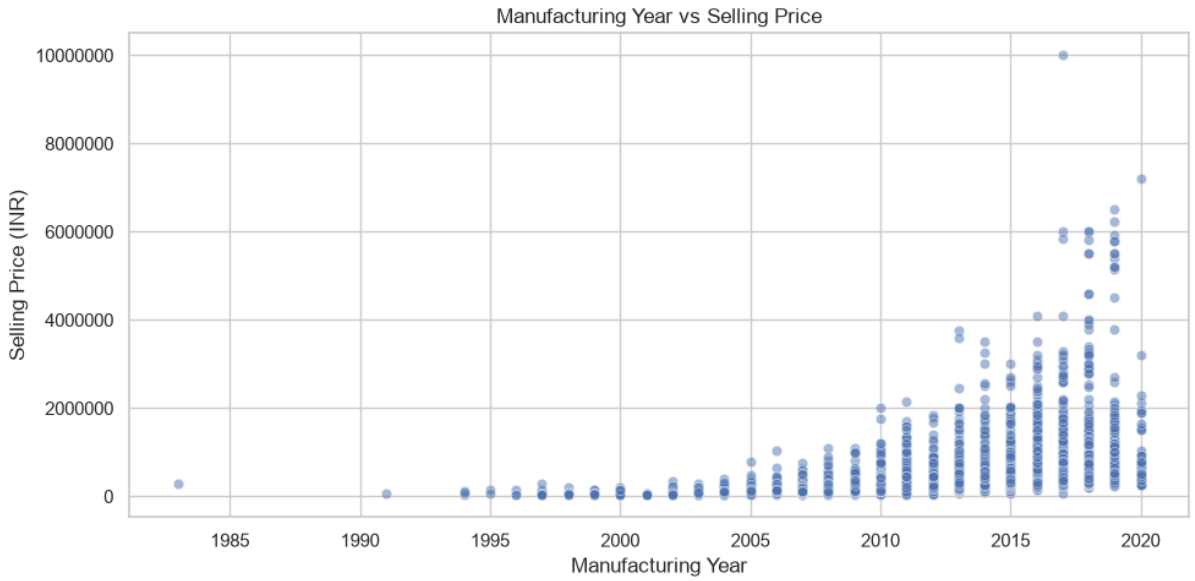


Figure 7.2: Manufacturing Year V/S Selling Price

7.3 Kilometres Driven vs Selling Price

- The relationship between kilometres driven and selling price was examined to understand how vehicle usage affects its resale value. The analysis indicates a general negative relationship between these two variables. Vehicles with lower kilometres driven tend to have higher selling prices, while vehicles with greater usage generally have lower selling prices.
- To examine this relationship, the vehicles were divided into five groups based on kilometre-driven quantiles. The median selling price decreased from approximately ₹5,64,000 in the lowest kilometre group to ₹3,25,000 in the highest kilometre group. The decline was more noticeable across the first four groups, while the difference between the final two groups was comparatively smaller.
- These observations suggest that kilometres driven is an important factor associated with used car selling prices. However, the relationship may also depend on manufacturing year, vehicle condition, and other specifications.

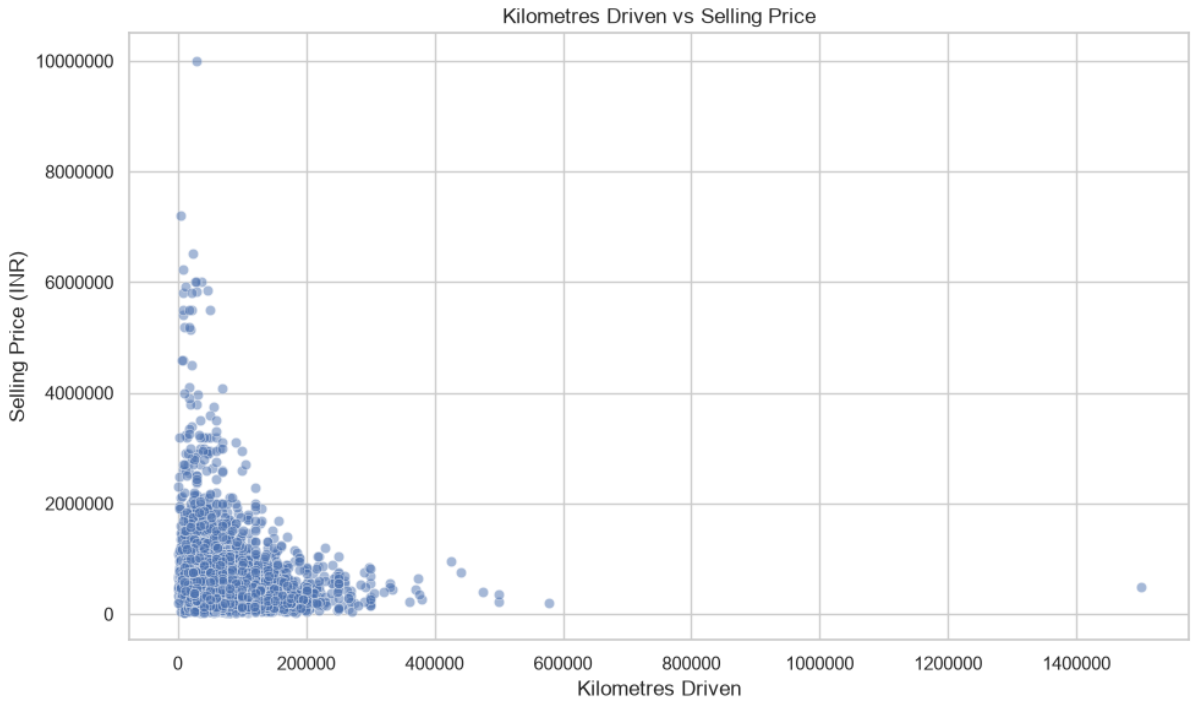


Figure 7.3: Kilometres Driven V/S Selling Price

7.4 Fuel Type vs Selling Price

- The relationship between fuel type and selling price was analysed to understand the price differences among vehicles using different fuel categories. The analysis showed that diesel vehicles had the highest average selling price of approximately ₹6,39,728, followed by petrol vehicles at approximately ₹3,75,632. CNG vehicles recorded an average selling price of approximately ₹3,00,500, while LPG vehicles had an average selling price of approximately ₹2,00,421.
- The dataset contained 3,755 diesel vehicles, 3,076 petrol vehicles, 56 CNG vehicles, and 38 LPG vehicles. Diesel and petrol vehicles formed the majority of the dataset, whereas CNG and LPG vehicles had comparatively fewer records.
- The results indicate differences in selling prices across fuel categories. However, these differences may also be influenced by other vehicle characteristics, such as manufacturing year, engine capacity, manufacturer, and model. Therefore, fuel type should be considered alongside other features when predicting used car prices.

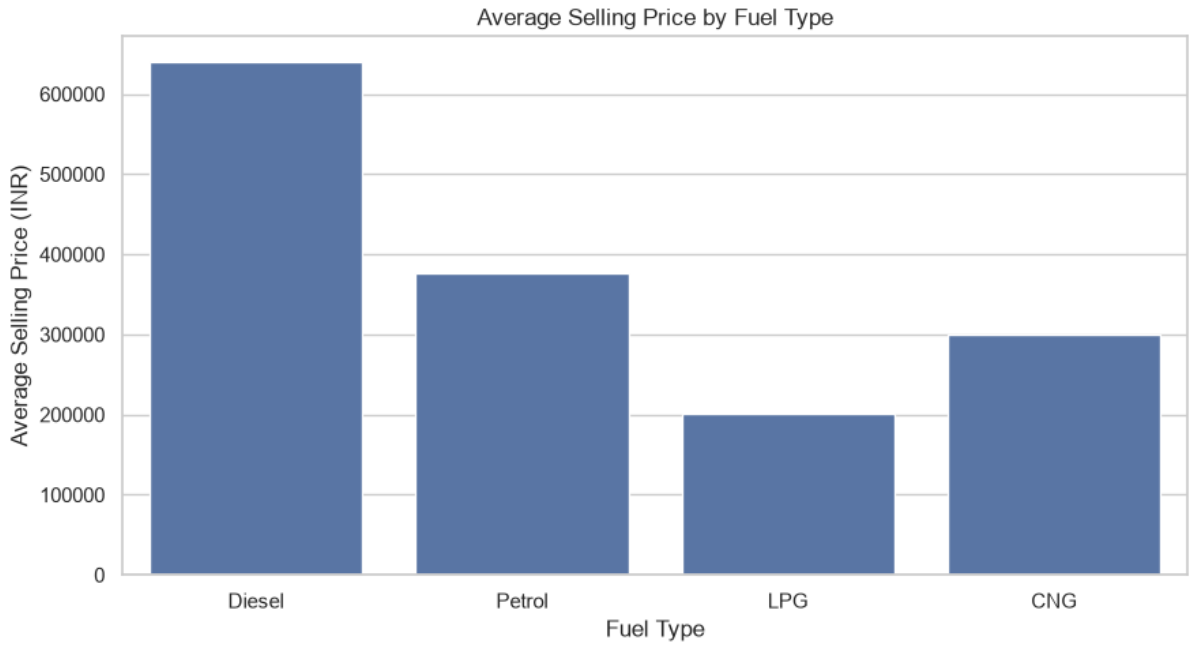


Figure 7.4: Average Selling Price By Fuel Type

7.5 Seller Type vs Selling Price

- The relationship between seller type and selling price was examined to understand whether vehicle prices differ according to the category of seller. The analysis showed that vehicles sold by dealers had an average selling price of approximately ₹9,26,722, while vehicles sold by Trustmark Dealers recorded an average of approximately ₹7,18,111. Vehicles sold by individual sellers had an average selling price of approximately ₹4,71,543.
- The dataset contained 6,217 vehicles sold by individuals, 681 vehicles sold by dealers, and 27 vehicles sold by Trustmark Dealers. Individual sellers represented the largest category in the dataset.
- The results suggest that dealer-listed vehicles tend to have higher selling prices than vehicles listed by individual sellers. However, this difference may also be influenced by vehicle specifications, manufacturer, model, and market segment. The Trustmark Dealer category has relatively few records, so its average should be interpreted cautiously.

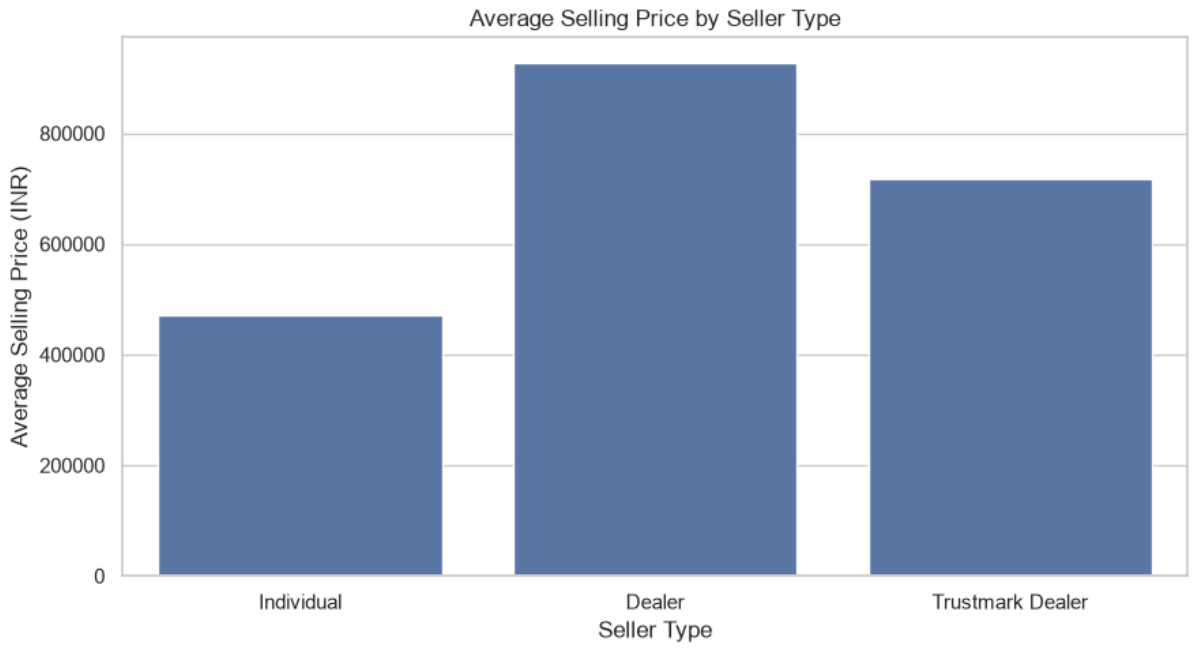


Figure 7.5: Average Selling Price By Seller Type

7.6 Transmission Type vs Selling Price

The relationship between transmission type and selling price was analysed to understand the price differences between automatic and manual vehicles.

The analysis shows that automatic vehicles have an average selling price of approximately ₹13,09,712, whereas manual vehicles have an average selling price of approximately ₹4,44,282. The median selling prices were ₹8,50,000 for automatic vehicles and ₹3,85,000 for manual vehicles.

The dataset contains 6,341 manual vehicles and 584 automatic vehicles, indicating that manual transmission vehicles are significantly more common.

The results suggest that automatic vehicles tend to have higher selling prices than manual vehicles. However, other factors, such as vehicle model, manufacturing year, manufacturer, and engine specifications, may also contribute to this difference.

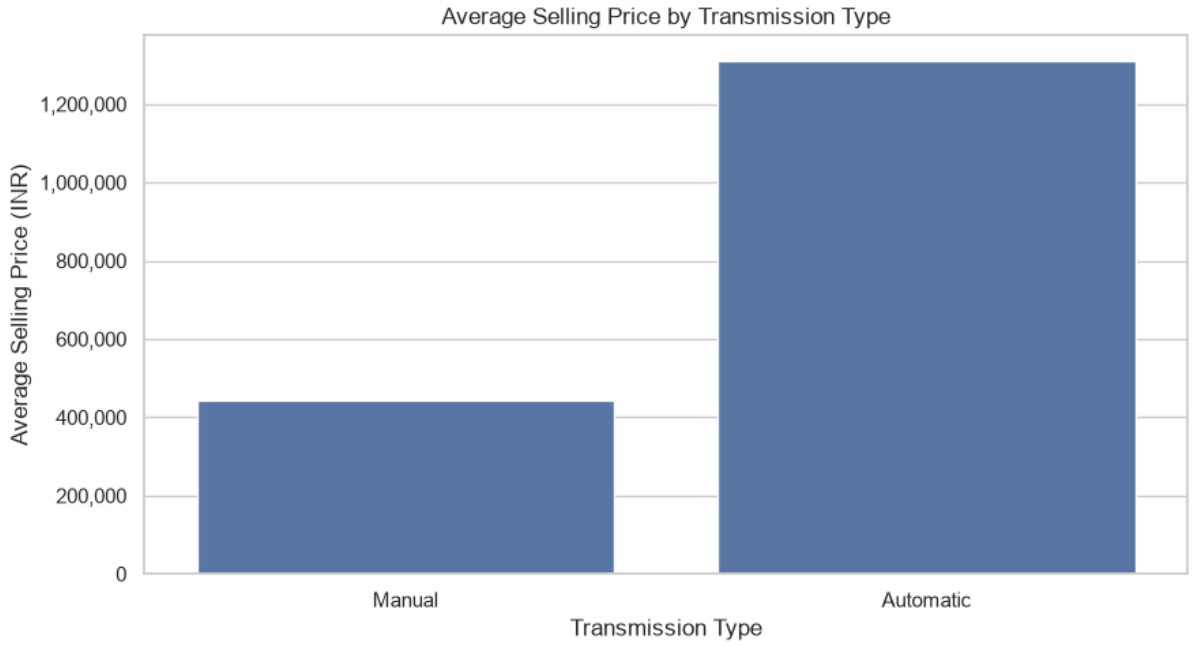


Figure 7.6: Average Selling Price By Transmission Type

7.7 Owner Type vs Selling Price

The relationship between ownership history and selling price was analysed to understand how the number of previous owners is associated with vehicle resale value. The results indicate that vehicles with fewer previous owners generally have higher selling prices compared to vehicles that have changed ownership multiple times.

First-owner vehicles recorded an average selling price of approximately ₹6,15,404, while second-owner vehicles recorded an average of approximately ₹3,85,614. Third-owner vehicles had an average selling price of approximately ₹2,81,251, whereas vehicles belonging to the fourth-and-above-owner category recorded an average of approximately ₹2,24,470.

The test-drive car category recorded an average selling price of approximately ₹44,03,800, but it contained only five records. Therefore, this category is too small to support a reliable general conclusion.

Overall, the analysis suggests that ownership history is associated with used car selling prices, with vehicles having fewer previous owners generally commanding higher prices.

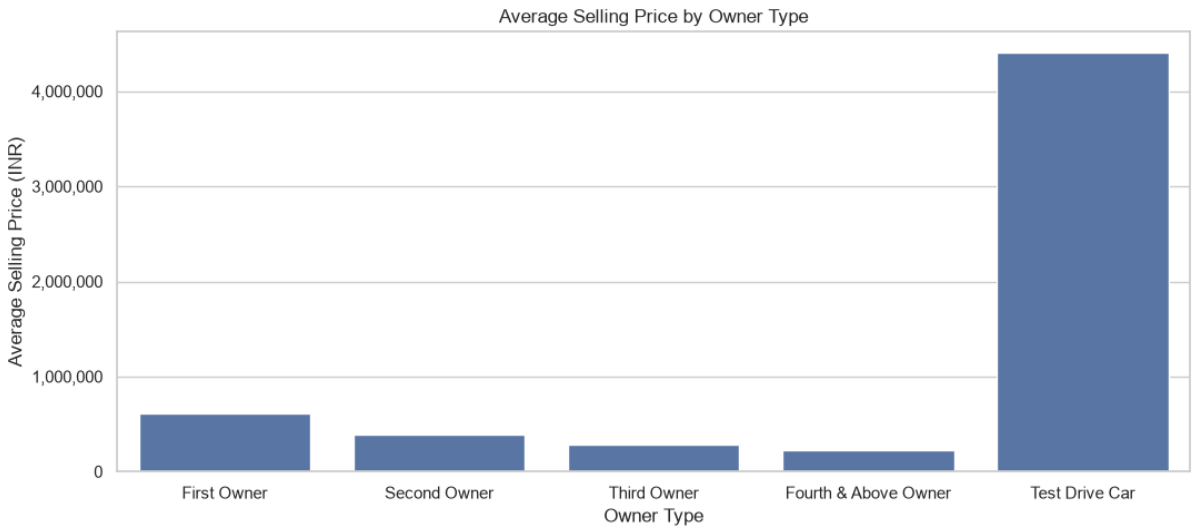


Figure 7.7: Average Selling Price By Owner Type

7.8 Correlation Analysis

Correlation analysis was performed to examine the strength and direction of relationships between numerical vehicle features and selling price. The results showed that maximum power had the strongest positive correlation with selling price, with a correlation coefficient of approximately 0.69, followed by torque at 0.63, engine capacity at 0.44, and manufacturing year at 0.43.

Kilometres driven and mileage showed weak negative correlations with selling price, with coefficients of approximately -0.19 and -0.11, respectively. Seating capacity showed a weak positive correlation of approximately 0.16.

The analysis also identified strong relationships between certain input features. Maximum power and torque had a correlation of approximately 0.84, indicating that vehicles with higher engine power often have higher torque values. Engine capacity also showed positive correlations with torque and maximum power.

These findings indicate that engine-related specifications and manufacturing year have relatively strong linear associations with selling price in this dataset. However, correlation does not establish causation, and the combined influence of multiple features is considered during machine learning model development.

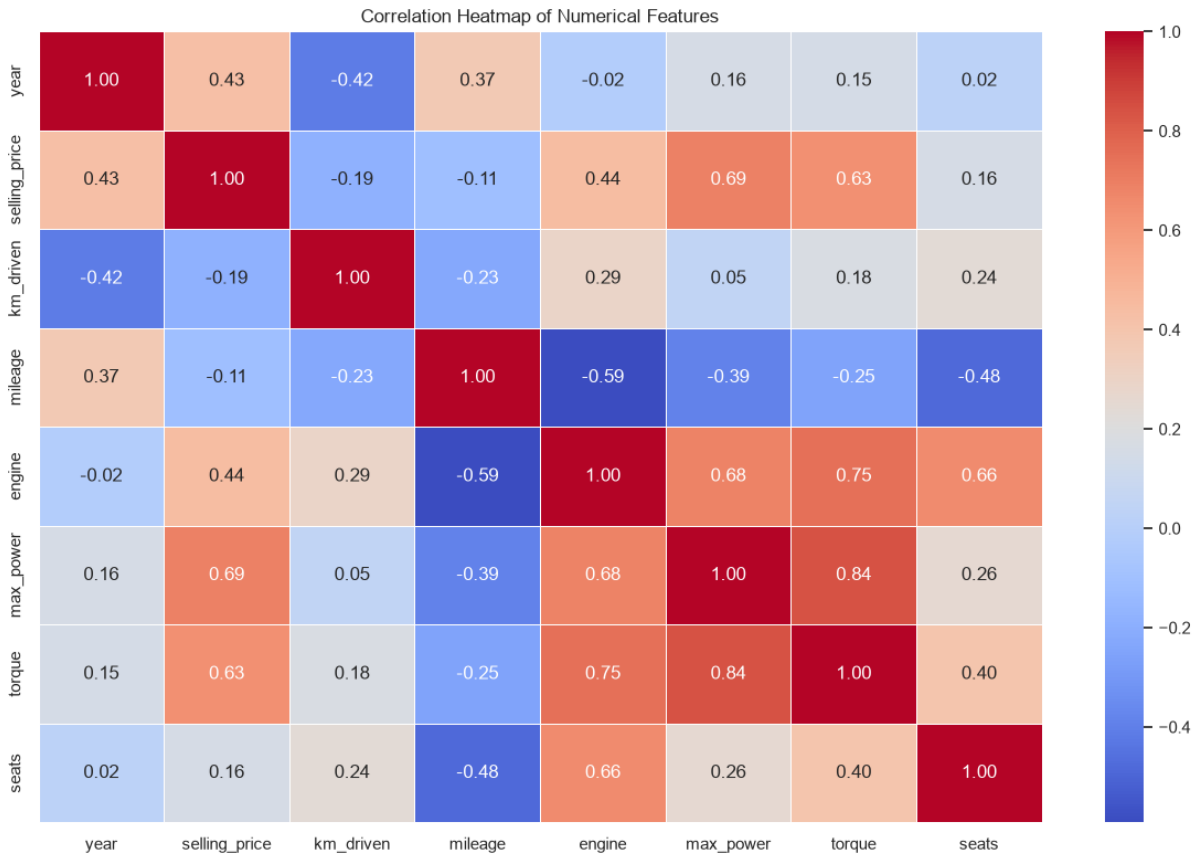


Figure 7.8: Correlation Heatmap of Numerical Features

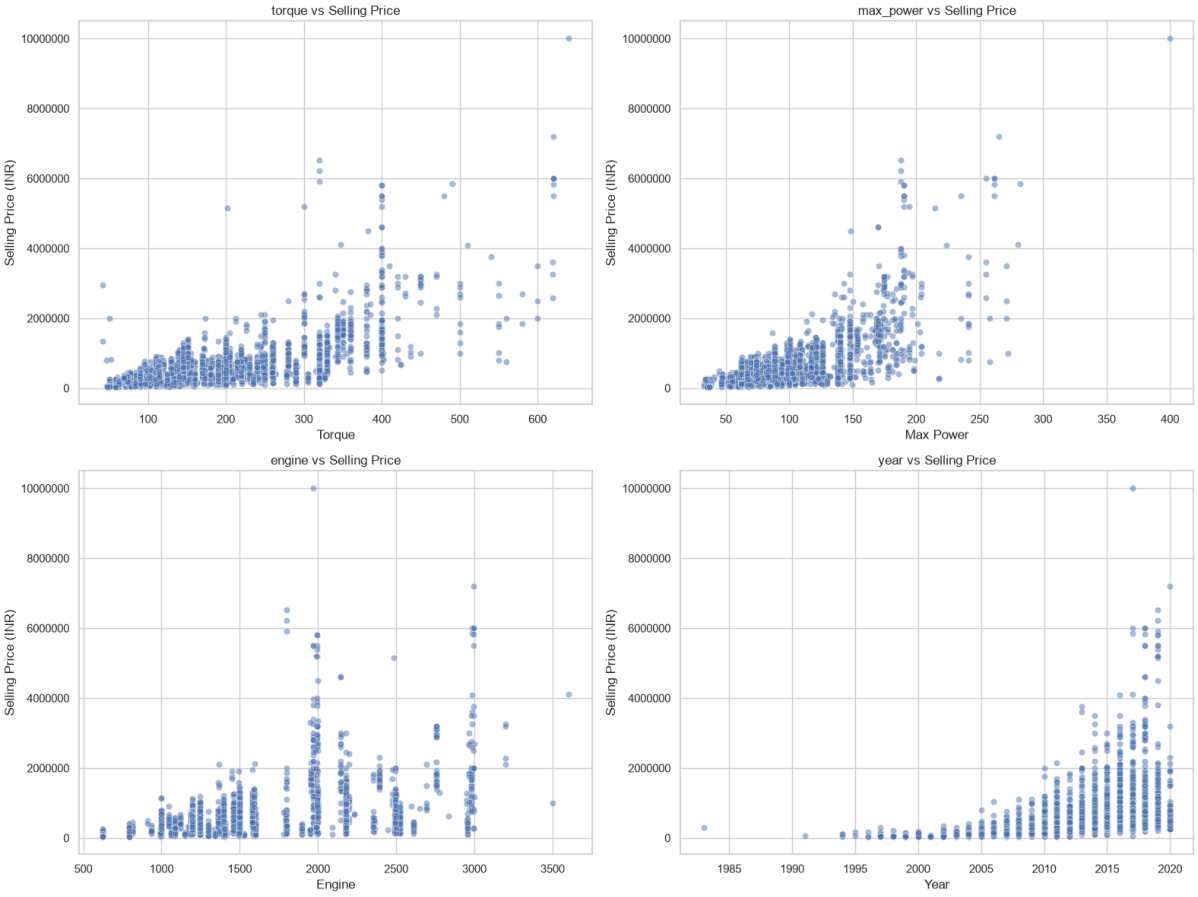


Figure 7.9: Visualise feature relationships

8. Model Development and Implementation

8.1 Feature and Target Selection

After completing the exploratory data analysis, the dataset was prepared for model development. The selling_price column was selected as the target variable because the main objective of the project is to predict the selling price of a used car. The remaining relevant attributes, including year, kilometres driven, fuel type, seller type, transmission, owner type, mileage, engine capacity, maximum power, torque, seating capacity, manufacturer, and mileage unit, were selected as input features. The name column was excluded because it contained a large number of unique car model names, which could increase the complexity of categorical encoding. The manufacturer information was retained as a separate feature.

8.2 Train-Test Split

The cleaned dataset was divided into training and testing subsets using an 80:20 ratio. The training dataset contained 5,540 records, while the testing dataset contained 1,385 records. The train_test_split() function from Scikit-learn was used with a random state of 42 to ensure reproducibility. The training data was used to develop the regression models, while the testing data was kept separate to evaluate their performance on previously unseen records.

8.3 Data Preprocessing

Before training the models, a preprocessing pipeline was created to handle numerical and categorical features separately. Missing numerical values were replaced using the median strategy, followed by standardisation using StandardScaler. Missing categorical values were filled using the most frequent value, and OneHotEncoder was applied to convert categorical variables into numerical representations. A ColumnTransformer was used to combine these preprocessing operations into a single workflow. After encoding, the training and testing datasets contained 53 transformed features. The preprocessing steps were fitted only on the training data to prevent data leakage and ensure a fair model evaluation.

8.4 Linear Regression Model

Linear Regression was implemented as the baseline regression model for predicting used car prices. It estimates the relationship between input features and the target variable by fitting a linear equation to the training data. This model provided an initial performance benchmark against which the more advanced regression algorithms could be compared. The trained model was evaluated using the test dataset to measure its prediction error and explanatory performance.

8.5 Decision Tree Regression Model

A Decision Tree Regressor was implemented to capture non-linear relationships between car attributes and their selling prices. The algorithm works by dividing the dataset into smaller groups based on feature conditions and generating predictions from the resulting decision structure. Unlike Linear Regression, it can model more complex patterns in the data without assuming a linear relationship between the input features and the target variable.

8.6 Random Forest Regression Model

A Random Forest Regressor was implemented to improve prediction performance by combining multiple decision trees. Each tree is trained using variations of the training data and selected features, and their predictions are combined to produce the final result. The initial Random Forest model was configured with 200 estimators, a random state of 42, and parallel processing enabled through n_jobs=-1. This ensemble approach helped capture complex relationships in the dataset and reduce the limitations of relying on a single decision tree.

8.7 Hyperparameter Tuning

Hyperparameter tuning was performed to identify a suitable configuration for the Random Forest model. RandomizedSearchCV from Scikit-learn was used to explore different combinations of parameters, including the number of estimators, maximum tree depth, minimum samples required for splitting, minimum samples per leaf, and the number of features considered during splitting. The search evaluated 20 parameter combinations using five-fold cross-validation, with negative mean absolute error as the scoring measure. The best configuration contained 200 estimators, no maximum depth restriction, a minimum split of 2, a minimum leaf size of 1, and a maximum feature setting of 0.8. The best cross-validation MAE was approximately ₹79,325.53.

8.8 Complete Machine Learning Pipeline

After comparing the initial and tuned Random Forest configurations, the original Random Forest model was selected for the final implementation because it achieved better overall performance across the test evaluation metrics. A complete Scikit-learn Pipeline was created by combining the preprocessing steps with the selected Random Forest Regressor. This allowed raw input data to pass through the required imputation, scaling, and encoding operations before generating a price prediction. The complete pipeline was trained on the training dataset and saved as used_car_price_model.joblib using Joblib. The saved model was subsequently loaded and tested to verify that it could generate predictions successfully.

9. Model Evaluation and Performance Comparison

9.1 Evaluation Metrics

The performance of the regression models was evaluated using three standard metrics: Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the R-squared (R^2) score. MAE measures the average absolute difference between the actual and predicted selling prices, making it useful for understanding the typical prediction error in rupees. RMSE measures the square root of the average squared prediction errors and gives greater weight to larger errors. The R^2 score indicates how much of the variation in the target variable is explained by the model. A lower MAE and RMSE indicate better predictive performance, while a higher R^2 score indicates a better fit.

9.2 Model Performance Comparison

Four model configurations were evaluated using the same held-out test dataset containing 1,385 records. Linear Regression achieved an MAE of ₹134,109.10, an RMSE of ₹263,647.10, and an R^2 score of 0.7152. Decision Tree Regression improved the results, achieving an MAE of ₹96,463.49, an RMSE of ₹196,234.42, and an R^2 score of 0.8422. The original Random Forest Regressor achieved an MAE of ₹71,381.00, an RMSE of ₹138,354.48, and an R^2 score of 0.9216. The tuned Random Forest achieved a slightly lower MAE of ₹71,183.91, but its RMSE increased to ₹139,522.43 and its R^2 score decreased to 0.9202.

9.3 Comparison of Evaluation Results

The comparison indicates that both Random Forest configurations performed considerably better than Linear Regression and Decision Tree Regression. The original Random Forest achieved the lowest RMSE and highest R^2 score among all evaluated models, while the tuned Random Forest achieved the lowest MAE. The differences between the two Random Forest configurations were relatively small. However, the original Random Forest provided better overall performance across the three evaluation metrics, particularly in terms of RMSE and R^2.

Model	MAE (₹)	RMSE (₹)	R^2 Score
Linear Regression	134,109.10	263,647.10	0.7152
Decision Tree Regression	96,463.49	196,234.42	0.8422
Original Random Forest	71,381.00	138,354.48	0.9216
Tuned Random Forest	71,183.91	139,522.43	0.9202

9.4 Final Model Selection

Based on the overall evaluation results, the original Random Forest Regressor was selected as the final model for the project. It achieved an R^2 score of 0.9216, indicating that it explained approximately 92.16% of the variation in selling prices on the test dataset. Its MAE of ₹71,381.00 represents the average absolute prediction error, while its RMSE of ₹138,354.48 reflects the influence of larger prediction errors. Although hyperparameter tuning slightly reduced the MAE, it did not improve the other two evaluation metrics. Therefore, the original Random Forest was retained as the final model.

9.5 Generalisation and Model Performance

The selected Random Forest model achieved a training R^2 score of 0.9864 and a testing R^2 score of 0.9216. The difference between these scores suggests some degree of overfitting, meaning that the model performs better on the data used for training than on unseen data. However, its test performance remained strong compared with the other evaluated models. The results should be interpreted in the context of the available dataset and test split, as actual used car prices may vary depending on location, vehicle condition, market demand, and other factors not represented in the dataset.

10. Model Prediction and Deployment

10.1 Saving the Trained Model

After selecting the original Random Forest Regressor as the final model, the complete machine learning pipeline was saved using the Joblib library. The saved pipeline includes both the preprocessing operations and the trained regression model. It was stored in the project's models directory with the filename used_car_price_model.joblib. Saving the complete pipeline ensures that the same preprocessing operations used during model training are automatically applied when new car details are provided for prediction.

10.2 Loading and Verifying the Model

The saved model was loaded again using Joblib to verify that it could be reused successfully without repeating the training process. The loaded object was confirmed to be a Scikit-learn Pipeline containing the preprocessing and prediction components. This verification demonstrated that the trained model had been saved correctly and was ready to process new input records.

10.3 Predicting the Selling Price of a New Car

To test the model with unseen input data, a sample car record was created using details such as manufacturing year, kilometres driven, fuel type, seller type, transmission, ownership history, mileage, engine capacity, maximum power, torque, seating capacity, manufacturer, and mileage unit. The sample represented a 2018 Maruti petrol car with 50,000 kilometres driven, manual transmission, and first-owner status. The trained pipeline processed these input features and generated a predicted selling price of approximately ₹5,01,389.97.

10.4 Prediction Output

The successful prediction demonstrated that the model could accept new car specifications and estimate their selling prices without requiring manual preprocessing. The complete pipeline automatically handled missing-value imputation, numerical scaling, and categorical encoding before generating the prediction. This makes the model reusable for future applications, such as a web-based used car price estimation system.

10.5 Model Deployment Possibilities

The saved machine learning pipeline can be integrated into a user-facing application through a backend framework such as Flask or FastAPI. A user could enter the specifications of a used car through a web interface, and the backend could pass those details to the saved model to generate an estimated selling price. Although a complete web deployment was not implemented as part of the current project, the saved pipeline provides a foundation for developing such an application in the future.

11. Conclusion and Future Scope

11.1 Conclusion

The project, Used Car Price Prediction Using Machine Learning Regression, focused on developing a machine learning model to estimate the selling price of used cars based on their specifications and characteristics. The CarDekho dataset, originally containing 8,128 records and 13 columns, was cleaned and transformed to obtain 6,925 records with 15 columns. Exploratory Data Analysis was performed to understand the distribution of selling prices and examine the relationships between vehicle attributes and their prices.

Four regression model configurations were developed and evaluated: Linear Regression, Decision Tree Regression, the original Random Forest Regressor, and the tuned Random Forest Regressor. Among these, the original Random Forest achieved the best overall performance, with an MAE of ₹71,381.00, an RMSE of ₹138,354.48, and an R^2 score of 0.9216 on the test dataset. The complete preprocessing and prediction pipeline was saved using Joblib and successfully tested by generating a selling price prediction of approximately ₹5,01,389.97 for a sample car.

Overall, the project demonstrated how machine learning regression techniques can be applied to used car price estimation. It also provided practical experience in data preprocessing, exploratory analysis, feature engineering, model development, hyperparameter tuning, evaluation, and model persistence.

11.2 Limitations

The model was developed using a historical dataset, so its predictions may not fully reflect current used car market conditions. Certain important factors, such as vehicle condition, accident history, service records, geographical location, insurance status, and current market demand, were not available as input features. The dataset also contained missing values in several vehicle specifications, which were handled through imputation. Additionally, the model was evaluated using a single train-test split, meaning that its performance may vary with different data samples. Therefore, the predictions should be treated as estimates rather than guaranteed market prices.

11.3 Future Scope

The project can be extended by collecting a larger and more recent dataset containing used car listings from different locations and market segments. Additional features such as vehicle condition, accident history, service records, registration location, and market demand could improve prediction quality. Further experimentation with advanced regression algorithms and ensemble techniques could also be performed to evaluate potential improvements. The saved machine learning pipeline could be integrated with a Flask or FastAPI backend and a React-based frontend to develop an interactive web application. Such an application would allow users to enter car specifications and receive an estimated selling price through a simple interface. The model could also be periodically retrained using updated market data to maintain its relevance over time.

12. References

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